

Workforce Attrition Patterns and Risk Hotspot Analysis at Palo Alto Networks

Executive Summary

Employee attrition is a critical challenge for organizations as it impacts productivity, increases recruitment costs, and leads to loss of valuable expertise. This project analyzes workforce attrition patterns at Palo Alto Networks using employee demographic, career, and workload-related information. The analysis identifies high-risk departments, job roles, demographic groups, and workplace factors contributing to employee turnover. The findings provide actionable recommendations to support data-driven workforce retention strategies.

1. Introduction

Employee retention is an important aspect of workforce management in modern organizations. High employee attrition can disrupt business operations, increase hiring expenses, and reduce organizational efficiency. Palo Alto Networks operates in a highly competitive and technology-driven environment where retaining skilled employees is essential.

This project aims to analyze workforce attrition patterns and identify key risk factors associated with employee turnover.

2. Problem Statement

The organization lacks clear visibility into:

- Which departments experience the highest attrition.
- Which job roles are most affected.
- Whether age, gender, or marital status influence attrition.
- The impact of overtime and business travel on employee turnover.
- The relationship between promotion delays and employee exits.

Without such insights, retention strategies remain reactive rather than proactive.

3. Objectives

The objectives of this project are:

- Calculate the overall attrition rate.
- Identify departments with high attrition.
- Analyze attrition across job roles.
- Examine demographic attrition patterns.
- Evaluate workload-related attrition factors.
- Provide recommendations to improve employee retention.

4. Dataset Description

The dataset contains workforce information for employees at Palo Alto Networks.

Dataset Summary

Metric	Value
Total Records	1470
Total Features	31
Missing Values	0
Duplicate Records	0

Target Variable

Attrition

- 0 = Employee Retained
- 1 = Employee Left

The dataset includes demographic, compensation, job role, career progression, and work-life balance attributes.

5. Data Cleaning and Validation

The following preprocessing steps were performed:

- Verified dataset dimensions and data types.
- Checked for missing values.
- Checked for duplicate records.
- Confirmed correct encoding of the Attrition variable.
- Created age-group categories for demographic analysis.

Cleaning Results

- Missing Values: 0
- Duplicate Records: 0
- Dataset Ready for Analysis

6. Exploratory Data Analysis

The analysis included:

- Attrition Distribution Analysis
- Department-wise Attrition Analysis
- Job Role Attrition Analysis
- Age Group Attrition Analysis
- Gender-wise Attrition Analysis
- Marital Status Analysis
- Overtime Impact Analysis
- Business Travel Impact Analysis
- Promotion Delay Analysis

The visualizations were developed using Matplotlib and Seaborn and are available in the project repository and Streamlit dashboard.

7. Key Findings

Overall Attrition

The organization recorded an attrition rate of 16.12%, indicating that employee turnover is a significant workforce concern.

Department Analysis

Research & Development and Sales departments experienced the highest attrition levels, indicating higher workforce instability in these functional areas.

Job Role Analysis

The highest attrition was observed among:

- Sales Executives
- Laboratory Technicians
- Research Scientists
- Sales Representatives

Demographic Analysis

Employees aged 26–35 experienced the highest attrition levels. Attrition decreased among older age groups.

Male employees recorded slightly higher attrition counts than female employees.

Single employees showed higher attrition compared to married and divorced employees.

Workload Analysis

Employees working overtime exhibited significantly higher attrition than employees who did not work overtime.

Employees who traveled frequently for business also showed increased attrition rates.

Career Progression Analysis

Employees experiencing longer periods without promotion were more likely to leave the organization, indicating the importance of career growth opportunities.

8. Recommendations

Based on the analysis, the following recommendations are proposed:

Recommendation 1

Reduce excessive overtime through improved workload management and staffing allocation.

Recommendation 2

Develop targeted retention programs for employees aged 26–35.

Recommendation 3

Provide clear career growth and promotion pathways.

Recommendation 4

Focus retention initiatives on Research & Development and Sales departments.

Recommendation 5

Monitor frequently traveling employees and provide additional support to reduce burnout.

Recommendation 6

Improve employee engagement and job satisfaction among high-risk job roles.

9. Conclusion

This project successfully analyzed workforce attrition patterns at Palo Alto Networks using employee demographic, career, and workload data.

The analysis revealed that attrition is concentrated among specific departments, job roles, and demographic groups. Overtime, frequent business travel, and promotion delays were identified as important factors influencing employee turnover.

The findings provide valuable insights that can help HR leadership implement targeted retention strategies, reduce turnover costs, and improve workforce stability through evidence-based decision-making.

10. References

- Unified Mentor Project Guidelines
- Palo Alto Networks Workforce Attrition Dataset
- Python Documentation
- Pandas Documentation
- Seaborn Documentation
- Streamlit Documentation